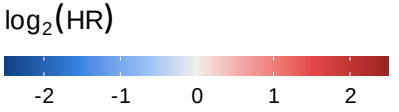
B1 Cause-specific hazard ratios by first-progression pattern (reference: Sys-TG)

All-cause: Cox HR; cause-specific: csHR (cause-specific Cox). Lunn–McNeil heterogeneity across IH/EH/Systemic HCC death: pattern × cause $\chi^2(16) = 60.5$, $P < 0.001$; site × cause $\chi^2(4) = 48.6$, $P < 0.001$; form × cause $\chi^2(4) = 7.4$, $P = 0.118$

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	0.54 (0.40–0.72)***	0.63 (0.43–0.92)*	0.75 (0.25–2.24)	0.28 (0.16–0.47)***	NE (sparse reference)
IH-NL	0.46 (0.34–0.62)***	0.57 (0.38–0.84)**	0.45 (0.14–1.40)	0.24 (0.14–0.42)***	NE (sparse reference)
IH-TG+NL	0.73 (0.52–1.02)	0.94 (0.61–1.46)	1.14 (0.32–3.99)	0.34 (0.16–0.68)**	NE (sparse reference)
EH-TG	0.84 (0.60–1.19)	0.90 (0.56–1.43)	2.01 (0.64–6.26)	0.48 (0.25–0.92)*	NE (sparse reference)
EH-NL	0.58 (0.43–0.78)***	0.48 (0.31–0.72)***	1.35 (0.46–3.95)	0.47 (0.28–0.78)**	NE (sparse reference)
EH-TG+NL	0.79 (0.48–1.29)	0.65 (0.31–1.33)	1.84 (0.39–8.67)	0.56 (0.23–1.36)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.78 (0.55–1.12)	0.59 (0.35–0.98)*	2.63 (0.83–8.33)	0.79 (0.43–1.45)	NE (sparse reference)
Sys-TG+NL	0.79 (0.58–1.07)	0.68 (0.44–1.03)	1.43 (0.47–4.35)	0.85 (0.50–1.44)	NE (sparse reference)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.397**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group. NE: <3 events in the cell or in the reference cell. * P<0.05, ** P<0.01, *** P<0.001 (unadjusted; BH-FDR q values in Table S-HR). Overall Wald test of the 8 pattern terms does not depend on the reference.